A SELECTION OF WORK / 2026

# Mohammad Ayaz Alam

**AI Engineer / Data Scientist / Product Builder**

## Building AI that works beyond the prototype.

I design and deliver practical AI systems, from reliable multi-agent workflows and production-ready applications to machine learning pipelines operating on large datasets. My experience connects industrial research, product engineering, and data science.

### SELECTED EVIDENCE

| 90M+ | 30,490 | 02 |
|---|---|---|
| Records handled in ML pipelines | Product-store demand series analyzed | Industrial workflows validated |

### WHERE I FOCUS

Agentic AI & LLM applications • Applied machine learning • Full-stack AI products

**RECENT EXPERIENCE** Zalion / Siemens Energy / BASF / IAB / Ilham Labs.ai

**Website** **LinkedIn** **GitHub** **Email**

01 / SELECTED WORK

# Products designed to be used.

Independent products built across the full software lifecycle, with an emphasis on clear interfaces, useful workflows, and dependable infrastructure.

## 01

AI WORKSPACE / PRODUCT

### Lattice Lab

A visual, collaborative workspace for exploring AI conversations as connected graphs instead of isolated chat threads. Users can branch questions, compare models, and merge the most valuable results while keeping context visible.

**BUILT** Multi-model comparison, graph-based conversations, authenticated shared workspaces, access roles, and PostgreSQL-backed persistence.

Next.js / FastAPI / PostgreSQL / Multi-model LLMs | **Explore Lattice Lab**

## 02

REALTIME / PRODUCT

### Air Lune

A multiplayer focus platform that makes deep work feel like a shared journey. Flight-inspired sessions create a more engaging setting for individual productivity and group study.

**BUILT** The full-stack focus experience, real-time collaboration architecture, and application services.

Next.js / FastAPI / LiveKit | **Visit Air Lune**

## 03

FULL STACK / PRODUCT

### WhenYouOpen.me

A scheduled-message platform for creating letters and memories that become available at a chosen future moment. The product combines a personal user experience with secure cloud-backed application infrastructure.

**BUILT** Scheduled message workflows, file storage integrations, and deployment infrastructure.

FastAPI / AWS S3 / Docker / Nginx | **Visit product**

**PERSONAL INITIATIVE / ILHAM LABS.AI**

Across these products, I take responsibility for architecture, application development, deployment, and the practical details that make software usable.

02 / SELECTED WORK

# Research into implementation.

Projects that connect model experimentation with robust systems, auditable workflows, and measurable evaluation.

## 04

INDUSTRIAL AI / THESIS

### Auditable Multi-Agent Systems

At Siemens Energy, designed a modular framework for industrial NLP applications. The architecture coordinates specialized agents, tool use, and persistent state, while controlling sensitive actions through human approval and audit trails.

**BUILT** LangGraph orchestration, PostgreSQL checkpointing, FastAPI interfaces, response streaming, and evaluation across Factory Assistance and Terminal Management System workflows.

LangGraph / LangChain / FastAPI / PostgreSQL

## 05

FORECASTING / APPLIED ML

### Retail Demand Forecasting

Created a 28-day retail sales forecasting workflow for 30,490 product-store time series using the M5 dataset. Compared classical and machine learning approaches with chronological validation.

**BUILT** Leakage-safe temporal features and LightGBM benchmarking. Achieved 0.7595 mean RMSSE, a 23.8% reduction compared with the weekly seasonal-naive baseline.

Python / LightGBM / ARIMA / SARIMA / Time series | **Repository**

## 06

AGENTS / AUTOMATION

### Multi-Agent Research System

Developed a research pipeline that organizes web retrieval, source reading, report writing, and critique as separate AI tasks.

**BUILT** Specialized search and reader agents, writer and critic stages, source-attributed reports, and Markdown exports in Streamlit.

LangChain / Tavily / Beautiful Soup / Streamlit | **Repository**

## 07

NLP / MODEL TRAINING

### Poetry Generation with RL

Fine-tuned GPT-2 (124M) on poetry and explored reinforcement learning as a way to guide text-generation characteristics.

**BUILT** Custom reward signals for rhyme, coherence, and creativity, with policy-gradient optimization after supervised fine-tuning.

GPT-2 / PyTorch / Hugging Face Transformers | **Model on Hugging Face**

03 / EXPERIENCE

# Applied across industries.

A progression through AI engineering, industrial research, consulting, and data-intensive machine learning.

**Mar - Jul 2026**
Munich, Germany

### Junior AI Engineer

**Zalion GmbH**

Built and integrated agentic workflows for AI-powered procurement applications.

- Connected specialist agents to conversational interfaces and built FastAPI services with Next.js components.
- Improved agent orchestration and contributed to Docker, AWS EC2, and CI/CD deployments.

**Jul 2025 - Feb 2026**
Erlangen, Germany

### Master's Thesis / Junior Data Scientist (NLP)

**Siemens Energy**

Developed a reusable multi-agent architecture for industrial assistants.

- Implemented persistent conversations, tool management, streaming APIs, and approval-based external actions.
- Validated the design in two industrial application scenarios.

**Nov 2024 - Feb 2026**
Nuremberg, Germany

### Research Assistant

**Institute for Employment Research (IAB)**

Worked on labor-market analytics and data-intensive job-offer ranking.

- Processed 90M+ records and developed feature-engineering pipelines for large-scale modeling.
- Applied XGBoost, Random Forest, SVM, and positive-unlabeled learning to predictive ranking problems.

**Apr - Jul 2025**
Ludwigshafen, Germany

### Data Consultant Intern

**BASF**

Built practical AI and software tools for internal business operations.

- Developed an AI-powered data validation agent with configurable checks, quality scoring, and reports.
- Created an internal event app with submission, voting, analytics, and access-control flows.

04 / CAPABILITIES

# The tools behind the work.

Broad enough to build complete products, focused on the technical depth needed for dependable AI and data systems.

### AI systems

Agentic AI, LangGraph, LangChain, LangSmith, MCP, RAG, tool workflows

### Machine learning & NLP

PyTorch, Scikit-learn, XGBoost, LightGBM, Hugging Face, Transformers

### Backend & application

Python, SQL, JavaScript, FastAPI, Django, Flask, Next.js, REST APIs

### Cloud & deployment

AWS EC2, S3, Lightsail, Docker, Git, CI/CD, Linux

### Databases & data

PostgreSQL, Firebase, Pandas, NumPy, data pipelines, query optimization

### Analytics & tooling

Power BI, Plotly, feature engineering, model evaluation, Flutter

## EDUCATION

**M.Sc. Data Science** 2022 - Present

Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany

**Bachelor of Computer Applications** 2018 - 2021

Guru Gobind Singh Indraprastha University, Delhi, India

## SELECTED CERTIFICATIONS

Hugging Face AI Agents Fundamentals / iNeuron Full Stack Data Analytics / iNeuron Deep Learning Foundation / Kaggle Intro to Machine Learning / freeCodeCamp Data Analysis with Python

# Let’s build something useful.

Open to conversations about AI engineering, applied machine learning, product development, and research collaborations.

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